

My Computer Science Modules at TU Berlin

Overview of modules completed or in progress at Technical University of Berlin, including content (hyperlinked), ECTS credits, and grades.

Module	Grade	Credits	Description
Introduction to AI	n.a.	6	Machine learning, optimization, linear regression
Stochastics for CS	2.7	9	Probabilities, distributions, statistics
Formal Languages	2.3	6	Automata theory, formal grammars
IT Security	n.g.	3	Cryptography, authentication
Distributed Apps	n.g.	6	Cloud computing, microservices
Algorithm Theory	2.0	6	Complexity, advanced algorithms
Computer Org	1.7	6	Architecture, CPU design
Calculus & LA	n.g.	12	Math foundations
CS Propaedeutics	n.g.	3	Intro to CS concepts
Programming	1.7	6	Trees, sorting, C
Discrete Math	2.7	6	Graph theory, number theory
Info Systems	1.3	6	Databases, SQL
Systems Prog	2.0	6	OS concepts, C
Algorithms	2.3	6	DFT, BFT, data structures
Logic	1.0	6	Propositional/predicate logic
Sci Computing	1.7	6	Numerical methods, FFT
Software Eng	3.0	6	Dev methodologies
Networks	3.0	6	Protocols, distributed systems
Predictability and Complexity	3.3	6	Turing machines, halting problem, P vs NP
Cognitive Algorithms	2.7	6	Classification, regression, deep learning
Data Integration: Algorithms and Systems	2.0	6	Schema Matching, data cleaning, data transformation
Data Integration and Large-scale Analysis	2.0	6	Data Cleaning, deduplication, cloud computing
Artificial Intelligence Fundamentals	2.3	6	Search Problems, machine Learning, KBS planning
AI: Quality and Usability	1.3	3	Usability, HCI, interfaces
Architecture of Machine Learning Systems	n.a.	6	Digital ethics, society, economy
Information Governance	1.7	6	Digital ethics, society, economy
Agent Technologies	1.7	6	Negotiation, Cooperation, Auctions
Distributed Systems	2.0	6	System Architecture, Coordination, Distributed Algorithms

Grades: 1.0 (best) - 4.0 (pass), increments of 0.3
n.a.: Currently taking or grade not yet available
n.g.: Passed but not graded numerically